

SUBMITTAL

Project

Kerr Stock Equipment

Date

Tuesday, July 18, 1905

Designer

applied@kerrcontrols.ca

Notes

KERR STOCK EQUIPMENT

Kerr Applied

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Prepared By: Kerr Applied 09:04AM

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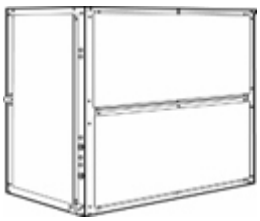
38AXQM08A0M5-0A0A0 + 40RLQA08A2A5-UA0A0

Submittal Cover Sheet
Unit Report
Performance Summary Report
Acoustic Summary
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Guide Specifications
Feature Sheet

Unit Report For 38AXQM08A0M5-0A0A0 + 40RLQA08A2A5-UA0A0

Project: Kerr Stock Equipment 10/25/2026
Prepared By: Kerr Applied 09:04AM





Outdoor Unit Parameters

Unit Quantity:	**1**	
Unit Model:	**38AXQ**	
Unit Size:	**7.5 Tons**	
Voltage:	**230-3-60**	V-Ph-Hz
No. of DX Circuits:	**1**	
No. of Stages:	**Dual Stage**	

Indoor Unit Parameters

Unit Model:	**40RLQ**	
Unit Size:	**7.5 Tons**	
No. of Coils:	**1**	
Voltage:	**230-3-60**	V-Ph-Hz

System Parameter

System Quantity:	**1**	
Refrigerant Type:	**PURONA**	
Compressor Quantity:	**1**	
Compressor Type:	**Hermetic**	
Std. Capacity Steps:	**67,100**	
Std. Min. Outdoor Temp(Cooling):	**35.0**	°F
No. of Outdoor fans:	**2**	

Outdoor Unit Dimensions and Weight

Unit Length:	**4' 11.4"**	
Unit Width:	**3' 9.9"**	
Unit Height:	**3' 6.4"**	
Unit Weight:	**523**	lb

Indoor Unit Dimensions and Weight

Unit Length:	**4' 1.0"**	
Unit Width:	**2' 4.2"**	
Unit Height:	**4' 8.1"**	
Unit Weight:	**385**	lb

Warranty Information Outdoor (Note: for US & Canada only)

First Year - Parts Only (Standard)

Warranty Information Indoor (Note: for US & Canada only)

First Year - Parts Only (Standard)

NOTE: Please see Warranty Catalog 808-218 for explanation of policies and ordering methods.

Ordering Information

Part Number	Description	Quantity
Base Unit - Outdoor		
38AXQM08A0M5-0A0A0		1
	Base Unit	
	Al/Cu with Louvered Hail Guard Condensing Coil	1
	Single Circuit/ Dual Stage Refrigerant Options	1
	Service Options - None	1
	Electrical Options - None	1
	Packaging Options - Standard	1
	Standard Electrical Mechanical Controls	1
	Refrig Circ/Compressor Staging - Single Circuit/Dual Stage	1
Base Unit - Indoor		
40RLQA08A2A5-UA0A0		1

	Base Unit	
	Standard/Medium Static (Ecoblue)	1
	Cabinet Paint - None	1
	2 Speed - Vane Axial Fan	1
	Electromechanical Unit Control Board	1

Performance Summary For 38AXQM08A0M5-0A0A0 + 40RLQA08A2A5-UA0A0

Project: Kerr Stock Equipment 10/25/2026
Prepared By: Kerr Applied 09:04AM

System:	**38AXQ008/40RLQ_08**	
No. of Circuits:	**Single Circuit**	
System Quantity:	**1**	
Altitude:	**0.0**	ft
EER @ ARI Conditions:	**11.2**	
IEER:	**15.3**	
COP @ AHRI Conditions:	**3.5**	
Suction Line Loss:	**1.4**	°F

System meets ASHRAE 90.1-2023 min IEER requirement.

This system has a combined part load efficiency (IEER) that meets or exceeds the US DOE 2023 minimum cooling efficiency standards.

Typical Liquid and Suction Line Sizing

Pipe Length	Liquid Line Size	Suction Line Size
0 - 25	1/2	1 1/8
26 - 50	1/2	1 1/8
51 - 75	1/2	1 1/8
76 - 100	1/2	1 1/8

Outdoor Unit Parameters

Unit Quantity:	**1**	
PartNumber:	**38AXQM08A0M5-0A0A0**	
Unit Model:	**38AXQ**	
Unit Size:	**7.5 Tons**	
No. of DX Circuits:	**1**	
Voltage:	**230-3-60**	V-Ph-Hz
Clg Ent Air DB:	**95.0**	°F
Htg Ent Air DB:	**47.0**	°F

Outdoor Electrical Data

Unit Voltage:	**230-3-60**	V-Ph-Hz
Unit MCA:	**37.0**	Amps
Unit MOCP:	**60.0**	Amps
Total Compressor Power of Unit:	**NA**	
Voltage Range Min:	**187**	V
Voltage Range Max:	**253**	V
Compressor RLA:	**26.3**	
Compressor LRA:	**191**	
Compressor Quantity:	**1**	
Fan Motors Qty:	**2**	
Fan Motor FLA:	**1.5**	Amps

Notice: Outdoor unit elect. data is based on 230-3-60

Control Panel SCCR: 5kA RMS at Rated Symmetrical Voltage

Indoor Unit Parameters

PartNumber:	**40RLQA08A2A5-UA0A0**	
Unit Model:	**40RLQ**	
Unit Size:	**7.5 Tons**	
Coil Rows:	**4**	
No. of Splits:	**1**	
Voltage:	**230-3-60**	V-Ph-Hz
Actual Airflow:	**3000.0**	CFM
Standard Airflow:	**3000.0**	CFM
Total Clg Cap.(Gross):	**92.8**	MBH
Sensible Clg Cap.(Gross):	**68.1**	MBH
Ent Air DB:	**80.0**	°F
Ent Air WB:	**67.0**	°F
Ent Enthalpy:	**31.44**	BTU/lb

Lvg Air DB:	**59.0**	°F
Lvg Air WB:	**57.3**	°F
Lvg Enthalpy:	**24.56**	BTU/lb
Coil Bypass Factor:	**NA**	

Heat Pump System

Htg HP Capacity:	**91.00**	MBH
Htg HP Integrated Capacity:	**91.00**	MBH
Heating HP Compressor Power:	**6.90**	kW
Htg Ent Air DB Temp:	**70.0**	°F
Htg Lvg Air DB:	**98.1**	°F
COP: (AHRI Conditions):	**3.5**	

Indoor Supply Fan

Indoor Unit External Static:	**0.60**	in wg
Economizer Loss:	**0.00**	in wg
Grille Loss:	**0.00**	in wg
Plenum Loss:	**0.00**	in wg
Dehumidification Loss:	**0.00**	in wg
Acc. Heating Loss:	**0.00**	in wg
Total Ext Static:	**0.60**	in wg
Fan Speed:	**1295**	RPM
Fan Power:	**0.77**	BHP
Fan Motor Max:	**2.40**	BHP
Fan Motor FLA:	**5.8**	Amps

Motor and Medium Static Drive Required.
Selected Medium Static Drive Range: 200 - 1788 RPM

Indoor Electrical Data

Unit Voltage:	**230-3-60**	V-Ph-Hz
Unit MCA:	**8.0**	Amps
Unit MOCP:	**15.0**	Amps

Notice: Indoor unit elect. data is based on 230-3-60

Acoustics

Sound Power Levels, db re 10E-12 Watts

A-Weighted	Outdoor Unit (dB)	Indoor Unit (dB,Ducted)
Total Level	82.4	88.4
63Hz	60.6	69.1
125Hz	69.6	75.2
250Hz	71.7	78.7
500Hz	77.1	83.1
1000Hz	77.7	82.3
2000Hz	73.5	81.5
4000Hz	71.2	77.3
8000Hz	64.3	NA
Sound Message	Sound for AXQ008	Sound for RLQ008

Acoustic Note:

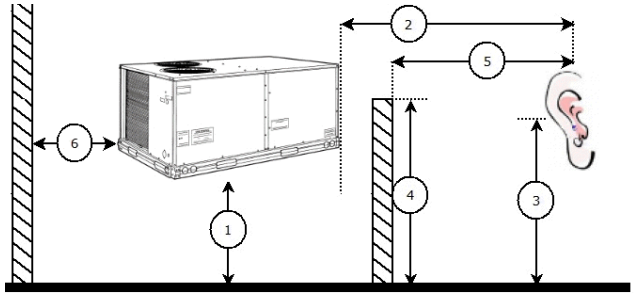
1. 38AXZ/D,38AUZ/D/Q units sound ratings are in accordance with AHRI 270-2008 - Sound Rating of Outdoor Unitary Equipment.
2. 40RL,40RF/40RU units sound ratings are in accordance with AHSRAE 1987 HVAC Systems and Applications handbook.
3. The acoustic center of the unit is located at the geometric center of the unit.

Performance Summary For 38AXQM08A0M5-0A0A0 + 40RLQA08A2A5-UA0A0

Project: Kerr Stock Equipment 10/25/2026
Prepared By: Kerr Applied 09:04AM

Advanced Acoustics Parameters

1. Unit height above ground: **1.0** ft
2. Horizontal distance from unit to receiver: **20.0** ft
3. Receiver height above ground: **5.7** ft
4. Height of obstruction: **0.0** ft
5. Horizontal dist. from obstruction to receiver: **0.0** ft
6. Horizontal dist. from unit to obstruction: **0.0** ft



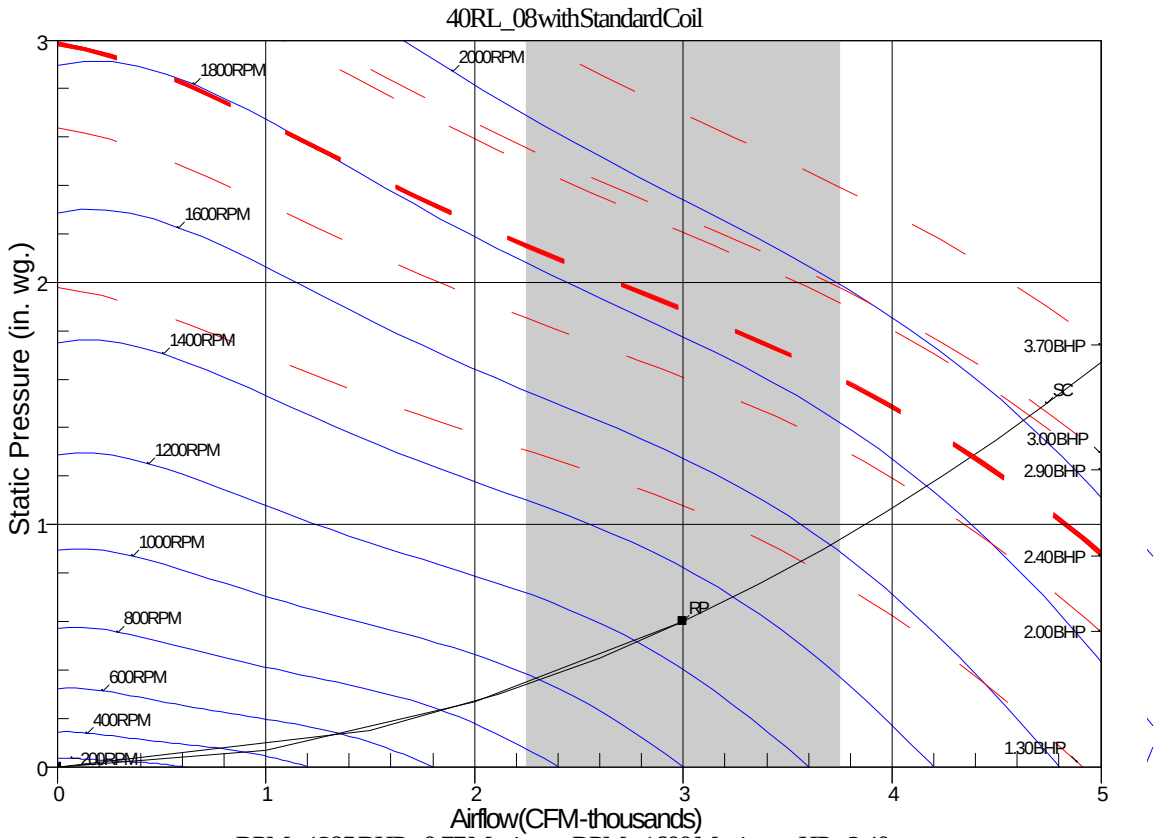
Detailed Acoustics Information

Octave Band Center Frequency, Hz	63	125	250	500	1k	2k	4k	8k	Overall
Sound Power Levels at Unit's Acoustic Center (Lw), dB	87	86	80	80	78	72	70	65	91
A-Wgted Sound Power Levels at Unit's Acoustic Center (LwA), dBA	61	70	72	77	78	74	71	64	82
Sound Press. Levels at Dist. Specified above (Lp), dB	62	61	56	56	53	48	46	41	66
A-Wgted Sound Press. Levels at Dist. Specified above (LpA), dBA	36	45	47	53	53	49	47	40	58

Calculation methods used in this program are patterned after the ASHRAE Guide; other ASHRAE Publications and the AHRI Acoustical Standards. While a very significant effort has been made to insure the technical accuracy of this program, it is assumed that the user is knowledgeable in the art of system sound estimation and is aware of the tolerances involved in real world acoustical estimation. This program makes certain assumptions as to the dominant sound sources and sound paths which may not always be appropriate to the real system being estimated. Because of this, no assurances can be offered that this software will always generate an accurate sound prediction from user supplied input data. If in doubt about the estimation of expected sound levels in a space, an Acoustical Engineer or a person with sound prediction expertise should be consulted.

Performance Summary For 38AXQM08A0M5-0A0A0 + 40RLQA08A2A5-UA0A0

Project: Kerr Stock Equipment 10/25/2026
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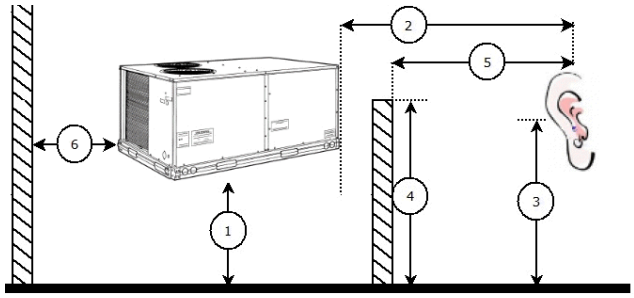
RPM=1295 BHP=0.77 Maximum RPM=1600 Maximum HP=2.40
Shaded Area-Recommended Operating Range
SC-System Curve RP-Rated Point MP-Minimum Point

Acoustic Summary For 38AXQM08A0M5-0A0A0 + 40RLQA08A2A5-UA0A0

Project: Kerr Stock Equipment 10/25/2026
Prepared By: Kerr Applied 09:04AM

Outdoor Unit Parameters:

Tag Name:**38AXQM08A0M5-0A0A0 + 40RLQA08A2A5-UA0A0**
Unit Model:**38AXQ**
Unit Size:**7.5 Tons**
System Type:**Dx Heat Pump**
Refrigerant Type:**PURONA**
Compressor Quantity:**1**
Compressor Type:**Hermetic**



Advanced Accoustics Parameters

1. Unit height above ground:**1.0** ft
2. Horizontal distance from unit to receiver:**20.0** ft
3. Receiver height above ground:**5.7** ft
4. Height of obstruction:**0.0** ft
5. Horizontal distance from obstruction to receiver:**0.0** ft
6. Horizontal distance from unit to obstruction:**0.0** ft

Detailed Acoustics Information

Octave Band Center Frequency, Hz	63	125	250	500	1k	2k	4k	8k	Overall
Sound Power Levels at Unit's Acoustic Center (Lw), dB	87	86	80	80	78	72	70	65	91
A-Wgted Sound Power Levels at Unit's Acoustic Center (LwA), dBA	61	70	72	77	78	74	71	64	82
Sound Press. Levels at Dist. Specified above (Lp), dB	62	61	56	56	53	48	46	41	66
A-Wgted Sound Press. Levels at Dist. Specified above (LpA), dBA	36	45	47	53	53	49	47	40	58

Calculation methods used in this program are patterned after the ASHRAE Guide; other ASHRAE Publications and the AHRI Acoustical Standards. While a very significant effort has been made to insure the technical accuracy of this program, it is assumed that the user is knowledgeable in the art of system sound estimation and is aware of the tolerances involved in real world acoustical estimation. This program makes certain assumptions as to the dominant sound sources and sound paths which may not always be appropriate to the real system being estimated. Because of this, no assurances can be offered that this software will always generate an accurate sound prediction from user supplied input data. If in doubt about the estimation of expected sound levels in a space, an Acoustical Engineer or a person with sound prediction expertise should be consulted.

Acoustic Note:

1. 38AXZ/D,38AUZ/D/Q units sound ratings are in accordance with AHRI 270-2008 - Sound Rating of Outdoor Unitary Equipment.
2. The acoustic center of the unit is located at the geometric center of the unit.
3. All estimated sound power levels, dB re 1 Picowatt should not be guaranteed or certified as being the actual sound power levels.

Acoustic Note:

1. Estimated Sound Power levels - dB re: 1 picowatt
2. Estimated Sound Pressure levels - dB re: 20 micropascal
3. Estimated sound levels given above are assumed to originate at the acoustic center of the unit. The acoustic center of the unit is located at the projection of the condensing unit's geometric center of its base.

4. Sound power levels shown above were determined in accordance with ARI standard 370 for large outdoor refrigeration and air conditioning equipment.

5. Calculation methods used in this program are patterned after the ASHRAE Guide; other ASHRAE Publications and the ARI Acoustical Standards. While a very significant effort has been made to insure the technical accuracy of this program, it is assumed that the user is knowledgeable in the art of system sound estimation and is aware of the tolerances involved in real world acoustical

estimation. This program makes certain assumptions as to the dominant sound sources and sound paths which may not always be appropriate to the real system being estimated. Because of this, no assurances can be offered that this software will always generate an accurate sound prediction from user supplied input data. If in doubt about the estimation of expected sound levels in a space, an Acoustical Engineer or a person with sound prediction expertise should be consulted.

Indoor Unit Parameters:

Tag Name: **38AXQM08A0M5-0A0A0 + 40RLQA08A2A5-UA0A0**
Unit Model: **40RLQ**
Unit Size: **7.5 Tons**
System Type: **Dx Heat Pump**
External Static Pressure: **0.60** in wg
Fan Speed: **1295** RPM
Fan BHP: **0.77** BHP

Detailed Acoustics Information

Octave Band Center Frequency, Hz	31	63	125	250	500	1k	2k	4k	8k	Total
Sound Power,dB	NA	95	91	87	86	82	80	76	NA	98
A-Weighted Sound Power, dBA	NA	69	75	79	83	82	82	77	NA	88

Acoustic Notes:

1. 40RU units sound ratings are in accordance with AHSRAE 1987 HVAC Systems and Applications handbook.
2. The acoustic center of the units is located at the geometric center of the unit.
3. All estimated sound power levels, dB re 1 Picowatt should not be guaranteed or certified as being the actual sound power levels.

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UNIT	ELECTRICAL CHARACTERISTICS	STD. UNIT WT.		CORNER A		CORNER B		CORNER C		CORNER D		CENTER OF GRAVITY			UNIT HEIGHT
		LBS.	KG.	LBS.	KG.	LBS.	KG.	LBS.	KG.	LBS.	KG.	X	Y	Z	H
38AXZ-07 (RTPF)	208/230-3-60,460-3-60,575-3-60	389	176	141	64	96	44	62	28	91	41	18 [457.2]	24 [609.6]	21 [533.4]	42-3/8 [1076.0]
38AXZ-08 (RTPF)	208/230-3-60,460-3-60,575-3-60	430	195	142	64	96	44	76	34	111	50	18 [457.2]	24 [609.6]	21 [533.4]	42-3/8 [1076.0]
38AXZ-12 (RTPF)	208/230-3-60,460-3-60,575-3-60	490	222	177	80	120	54	78	35	114	52	18 [457.2]	24 [609.6]	24 [609.6]	50-3/8 [1279.2]
38AXZ-14 (RTPF)	208/230-3-60,460-3-60,575-3-60	598	271	195	88	142	64	110	50	151	68	20 [508.0]	25 [635.0]	24 [609.6]	50-3/8 [1279.2]
38AXD-12 (RTPF)	208/230-3-60,460-3-60,575-3-60	516	234	185	84	117	53	88	38	131	59	19 [482.6]	23 [584.2]	24 [609.6]	50-3/8 [1279.2]
38AXD-14 (RTPF)	208/230-3-60,460-3-60,575-3-60	654	297	214	97	155	70	120	54	165	75	20 [508.0]	25 [635.0]	24 [609.6]	50-3/8 [1279.2]
38AXQ-07 (RTPF)	208/230-3-60,460-3-60,575-3-60	444	201	154	61	97	44	90	41	123	56	22 [558.8]	25 [635.0]	13 [330.2]	42-3/8 [1076.0]
38AXQ-08 (RTPF)	208/230-3-60,460-3-60,575-3-60	525	237	174	79	116	54	96	44	135	61	21 [533.4]	24 [609.6]	23 [584.2]	50-3/8 [1279.2]
38AXQ-12 (RTPF)	208/230-3-60,460-3-60,575-3-60	575	261	186	84	126	57	106	48	157	71	21 [533.4]	24 [609.6]	23 [584.2]	50-3/8 [1279.2]

NOTES:

1. MINIMUM CLEARANCE (LOCAL CODES OR JURISDICTION MAY PREVAIL):
   A. BOTTOM TO COMBUSTIBLE SURFACES: 0 INCHES.
   B. OUTDOOR COIL, FOR PROPER AIR FLOW: 36 INCHES ONE SIDE, 12 INCHES THE OTHER. THE SIDE GETTING THE GREATER CLEARANCE IS OPTIONAL. STANDARD CLEARANCES ON REMAINING TWO SIDES.
   C. OVERHEAD: 60 INCHES, TO ASSURE PROPER OUTDOOR FAN OPERATION.
   D. BETWEEN UNITS: CONTROL BOX SIDE, 42 INCHES PER NEC.
   E. BETWEEN UNIT AND UNGROUNDED SURFACES: CONTROL BOX SIDE, 36 INCHES PER NEC.
   F. BETWEEN UNIT AND BLOCK OR CONCRETE WALLS AND OTHER GROUNDED SURFACES: CONTROL BOX SIDE, 42 INCHES PER NEC.
2. WITH EXCEPTION OF THE CLEARANCE FOR THE OUTDOOR COIL AS STATED IN NOTE 1B, A REMOVABLE FENCE OR BARRICADE REQUIRES NO CLEARANCE.
3. UNITS MAY BE INSTALLED ON COMBUSTIBLE FLOORS MADE FROM WOOD OR CLASS A, B OR C ROOF COVERING MATERIAL.

ITC CLASSIFICATION	SHEET	DATE	SUPERCEDES	CONDENSING UNIT		REV
U.S. ECCN:NSR	1 OF 2	7/15/24	-	38AXZ/ 38AXD/ 38AXQ-07/08/12/14	38AU002464	-

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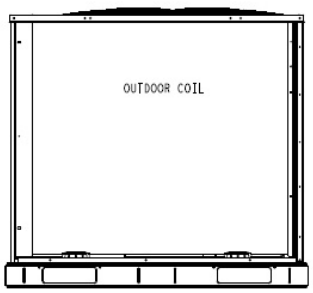
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CG CENTER OF GRAVITY

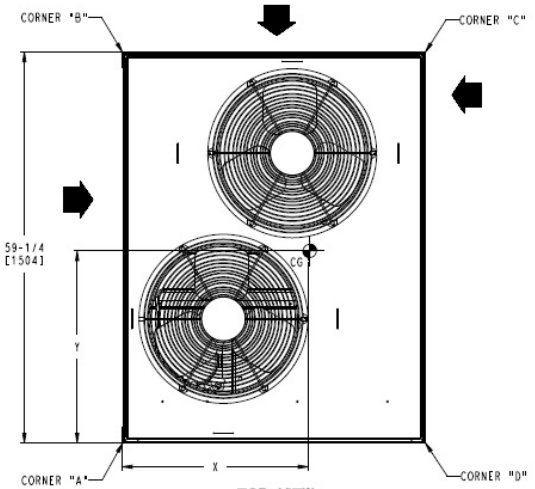
DIRECTION OF AIR FLOW

DIMENSIONS IN [ ] ARE IN MM

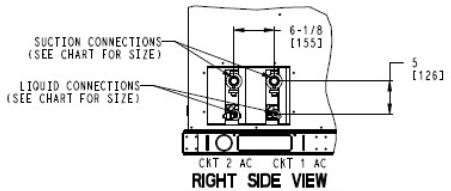
SERVICE VALVE CONNECTIONS			QTY
UNIT	SUCTION	LIQUID	
38AXZ07	1-1/8 [28.6]	3/8 [9.5]	1
38AXZ08	1-1/8 [28.6]	1/2 [12.7]	1
38AXZ12	1-3/8 [34.9]	1/2 [12.7]	1
38AXZ14	1-3/8 [34.9]	5/8 [15.9]	1
38AXD12	1-1/8 [28.6]	3/8 [9.5]	2
38AXD14	1-3/8 [34.9]	1/2 [12.7]	2
38AXQ07	1-1/8 [28.6]	3/8 [9.5]	1
38AXQ08	1-1/8 [28.6]	1/2 [12.7]	1
38AXQ12	1-3/8 [34.9]	1/2 [12.7]	1



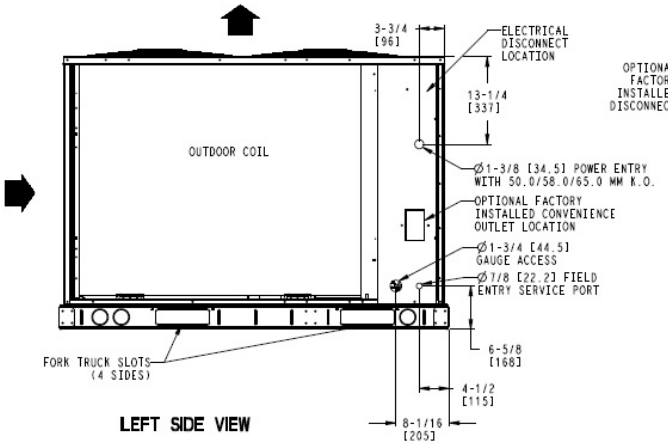
REAR VIEW



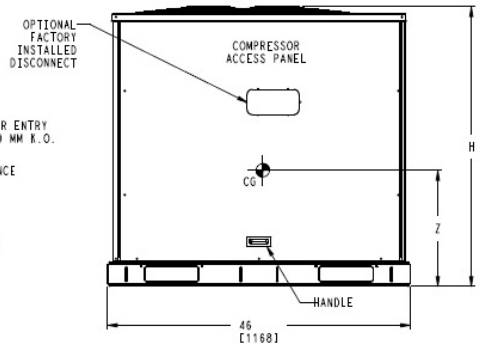
TOP VIEW



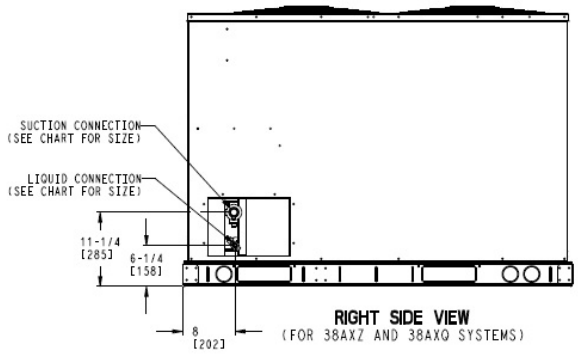
RIGHT SIDE VIEW
(FOR 38AXD SYSTEMS)



LEFT SIDE VIEW



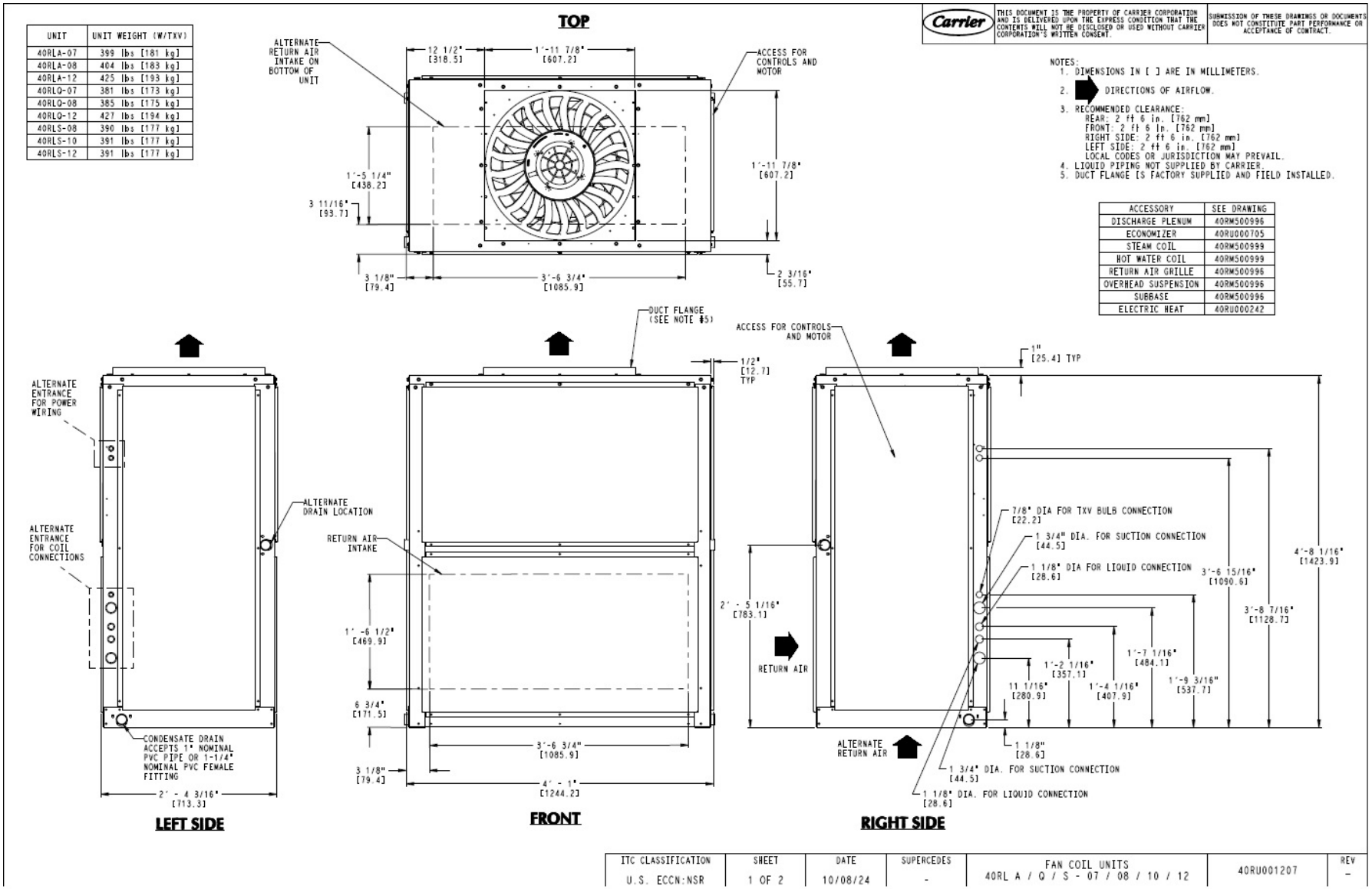
FRONT VIEW



RIGHT SIDE VIEW
(FOR 38AXZ AND 38AXQ SYSTEMS)

ITC CLASSIFICATION	SHEET	DATE	SUPERCEDES			REV
U.S. ECCN:NSR	2 OF 2	7/15/24	-	CONDENSING UNIT 38AXZ/ 38AXD/ 38AXQ-07/08/12/14	38AU002464	-

UNIT	UNIT WEIGHT (W/TXV)
40RLA-07	399 lbs [181 kg]
40RLA-08	404 lbs [183 kg]
40RLA-12	425 lbs [193 kg]
40RLQ-07	381 lbs [173 kg]
40RLQ-08	385 lbs [175 kg]
40RLQ-12	427 lbs [194 kg]
40RLS-08	390 lbs [177 kg]
40RLS-10	391 lbs [177 kg]
40RLS-12	391 lbs [177 kg]



ACCESSORY	SEE DRAWING
DISCHARGE PLENUM	40RM500996
ECONOMIZER	40RU000705
STEAM COIL	40RM500999
HOT WATER COIL	40RM500999
RETURN AIR GRILLE	40RM500996
OVERHEAD SUSPENSION	40RM500996
SUBBASE	40RM500996
ELECTRIC HEAT	40RU000242

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ITC CLASSIFICATION	SHEET	DATE	SUPERCEDES			REV
U.S. ECCN:NSR	1 OF 2	10/08/24	-	FAN COIL UNITS 40RL A / Q / S - 07 / 08 / 10 / 12	40RU001207	-

Commercial Air-Cooled Condensing Units HVAC Guide Specifications

Size Range: **6 to 20 Tons**

Carrier Model Numbers: **38AXQ, Single Circuit (07-12 Models) 38AXQ, Dual Circuit (16-25 Models)**

GENERAL

- SYSTEM DESCRIPTION

  Outdoor-mounted, electrically controlled, air-cooled split system heat pump suitable for on-the-ground or rooftop installation. Unit shall consist of a scroll air-conditioning compressor assembly, an air-cooled coil, propeller-type condenser fans, and a control box. Unit shall discharge supply air upward as shown on contract drawings. Unit shall function as the outdoor component of an air to air electric heat pump system.

- QUALITY ASSURANCE
- Unit shall be rated in accordance with AHRI Standard 340/360.
- Unit construction shall be designed to conform to ASHRAE 15.
- Unit shall be constructed in accordance with UL Standard 60335-1 and 60335-2-40, including testing to withstand rain and shall carry the UL and UL, Canada label.
- Unit cabinet shall be capable of withstanding 500-hour salt spray exposure per ASTM B117 (scribed specimen).
- Air-cooled condenser coils for hermetic scroll compressor units 38AXQ shall be leak tested at 150 psig, and pressure tested at 650 psig.
- Unit shall be manufactured in a facility registered to ISO 9001:2015 manufacturing quality standard.
- DELIVERY, STORAGE, AND HANDLING

  Unit shall be shipped as single package only, and shall be stored and handled according to unit manufacturer’s recommendations.

- WARRANTY (FOR INCLUSION BY SPECIFYING ENGINEER.)

Products

- EQUIPMENT
- General:

  Factory-assembled, single piece, air-cooled heat pump. Contained within the unit enclosure shall be all factory wiring, piping, controls, compressor, holding charge, and special features required prior to field start-up.

  Unit shall use Puron Advance™ (R-454B) refrigerant.

- Unit Cabinet:
  - Unit cabinet shall be constructed of galvanized steel, bonderized and coated with a prepainted baked enamel finish.
  - A heavy-gauge roll-formed perimeter base rail with forklift slots and lifting holes shall be provided to facilitate rigging.

- Condenser Fans:
  - Condenser fans shall be direct driven, propeller type, discharging air vertically upward.
  - Fan blades shall be balanced.
  - Condenser fan discharge openings shall be equipped with PVC-coated steel wire safety guards.
  - Condenser fan and motor shaft shall be corrosion resistant.
- Compressor:
  - Compressor shall be of the hermetic scroll type.
  - Compressor shall be mounted on rubber grommets.
  - Compressors shall include overload protection.
  - Compressors shall be equipped with a crankcase heater.
  - Compressor shall be equipped with internal high pressure and high temperature protection.
- Condenser Coils:
  - Standard Aluminum fin - Copper Tube Coils:
    - Standard evaporator and condenser coils shall have aluminum lanced plate fins mechanically bonded to seamless internally grooved copper tubes with all joints brazed.
    - Evaporator coils shall be leak tested to 150 psig, pressure tested to 450 psig, and qualified to UL 60335-2-40 burst test at 1775 psig.
    - Condenser coils shall be leak tested to 150 psig, pressure tested to 650 psig, and qualified to UL 60335-2-40 burst test at 1980 psig.
  - Optional copper-fin evaporator and condenser coils:
    - Shall be constructed of copper fins mechanically bonded to copper tubes and copper tube sheets.
    - Galvanized steel tube sheets shall not be acceptable.
    - A polymer strip shall prevent coil assembly from contacting the sheet metal coil pan to minimize potential for galvanic corrosion between coil and pan.
  - Optional e-coated aluminum-fin evaporator and condenser coils:
    - Shall have a flexible epoxy polymer coating uniformly applied to all coil surface areas without material bridging between fins.
    - Coating process shall ensure complete coil encapsulation of tubes, fins and headers.
    - Color shall be high gloss black with gloss per ASTM D523-89.
    - Uniform dry film thickness from 0.8 to 1.2 mil on all surface areas including fin edges.
    - Superior hardness characteristics of 2H per ASTM D3363-92A and cross-hatch adhesion of 4B-5B per ASTM D3359-93.
    - Impact resistance shall be up to 160 in.-lb (ASTM D2794-93).
    - Humidity and water immersion resistance shall be up to minimum 1000 and 250 hours respectively (ASTM D2247-92 and ASTM D870-92).
    - Corrosion durability shall be confirmed through testing to be no less than 1000 hours salt spray

per ASTM B117-90.

- Refrigeration Components:

  Refrigeration circuit components shall include liquid line service valve, suction line service valve, a full charge of compressor oil, and a partial holding charge of refrigerant.

- Controls and Safeties:
  - Minimum control functions shall include:
    - Control wire terminal blocks.
    - Compressor lockout on auto-reset safety until reset from thermostat.
    - Each unit shall utilize the Comfort Alert™ Diagnostic Board that provides:
      - System Pressure Trip fault code indication
      - Short Cycling fault code indication
      - Locked Rotor fault code indication
      - Open Circuit fault code indication
      - Reverse Phase 3 fault code indication
      - Welded Contactor fault code indication
      - Low Voltage fault code indication
      - Anti-short cycle protection
      - Phase reversal protection
  - Minimum safety devices which are equipped with automatic reset (after resetting first at thermostat), shall include:
    - High discharge pressure cutout.
    - Low pressure cutout.
  - Designed to integrate with A2L Refrigerant Leak detection system onboard 40RLQ products to enable unit shutdown in event of a leak.
- Operating Characteristics:
  - The capacity of the condensing unit shall meet or exceed _____ Btuh at a suction temperature of _____ °F/C. The power consumption at full load shall not exceed _____ kW.
  - The combination of the condensing unit and the evaporator or fan coil unit shall have a total net cooling capacity of _____ Btuh or greater at conditions of _____ cfm entering-air temperature at the evaporator at _____ °F/C wet bulb and _____ °F/C dry bulb, and air entering the condensing unit at _____ °F/C.
  - The system shall have an IEER of _____ Btuh/Watt or greater at standard AHRI conditions.
  - Standard unit shall be capable of operating up to 125°F (52°C) and down to 35°F (2°C)
- Electrical Requirements:
  - Nominal unit electrical characteristics shall be _____ v, 3-ph, _____ Hz. The unit shall be capable of satisfactory operation within voltage limits of _____ v to _____ v.
  - Unit electrical power shall be single-point connection.

- Unit control circuit shall contain a 24-v transformer for unit control.

- Special Features:
  - Low-Ambient Temperature Control:

    A low-ambient temperature control shall be available as a factory-installed option or as a field-installed accessory. This low-ambient control shall regulate speed of the condenser-fan motors in response to the saturated condensing temperature of the unit. The control shall maintain correct condensing pressure at outdoor temperatures down to –20°F (–29°C).

    Convenience Outlet:

    Poweredconvenience outlet.

    (1.) Outlet shall be powered from main line power to the rooftop unit.

    (2.) Outlet shall be powered from line side or load side of disconnect by installing contractor, as required by code. If outlet is powered from load side of disconnect, unit electrical ratings shall be UL certified and rated for additional outlet amperage.

    (3.) Outlet shall be factory-installed and internally mounted with easily accessible 115-v female receptacle.

    (4.) Outlet shall include 15 amp GFI receptacles with independent fuse protection.

    (5.) Voltage required to operate convenience outlet shall be provided by a factory-installed step-down transformer.

    (6.) Outlet shall be accessible from outside the unit.

    (7.) Outlet shall include a field-installed “Wet in Use” cover.

    Non-Powered convenience outlet.

    (1.) Outlet shall be powered from a separate 115/120v power source.

    (2.) A transformer shall not be included.

    (3.) Outlet shall be factory-installed and internally mounted with easily accessible 115-v female receptacle.

    (4.) Outlet shall include 15 amp GFI receptacles with independent fuse protection.

    (5.) Outlet shall be accessible from outside the unit.

    (6.) Outlet shall include a field-installed “Wet in Use” cover.

  - Unit-Mounted, Non-Fused Disconnect Switch:

    Switch shall be factory-installed and internally mounted. NEC and UL-approved non-fused switch shall provide unit power shutoff. Switch shall be accessible from outside the unit and shall provide power off lockout capability. Non-fused disconnect cannot be used when unit MOCP electrical rating exceeds 80 amps.

  - Thermostat Controls:
    - Programmable multi-stage thermostat shall have 7-day clock, holiday scheduling, large backlit display, remote sensor capability, and Title 24 compliance.
    - Commercial Electronic Thermostat shall have 7-day time clock, auto-changeover, multi-stage capability, and large LCD (liquid crystal display) temperature display.

  - Louvered Hail Guard Package:
    Louvered hail guard package shall protect coils against damage from hail and other flying debris.

6. Condenser Coil Grille:

   Grille shall add decorative appearance to unit and protect condenser coil from large objects and vandalism.

Commercial Packaged Air-Handling Unit

HVAC Guide Specifications

Size Range: **2,400 to 4,000 Cfm, Nominal Airflow, 6 to 10 Tons, Nominal Cooling**

Carrier Model Numbers: **40RLQ (Heat Pump Coil)**

General

- System Description
- Indoor, packaged air-handling unit for use in commercial heat pump split systems. Unit shall have a multi-position design and shall be capable of horizontal or vertical installation on a floor or in a ceiling, with or without ductwork. (Only vertical units are to be applied without ductwork.) Unit shall function as the indoor component of an air-to-air electric heat pump system.
- Unit with direct-expansion coil shall use Puron Advance™ (R-454B) in a refrigerant circuit with a matching air-cooled condensing unit.
- Quality Assurance
- Coils shall be designed and tested in accordance with ASHRAE 15 Safety Code for Mechanical Refrigeration (U.S.A.)
- Unit shall be constructed in accordance with UL Standard 60335-1 and 60335-2-40, and shall carry the UL and UL, Canada, labels.
- Unit insulation and adhesive shall comply with NFPA-90A (U.S.A.) requirements for flame spread and smoke generation.
- Unit shall be manufactured in a facility registered to the ISO 9001 manufacturing quality standard.
- Direct-expansion coils shall be burst and leak tested at 435 psi.
- Delivery And Storage
- Units shall be stored and handled per manufacturer’s recommendations.
- Safeties

A. A2L Refrigerant Leak Dissipation System (Electromechanical)

- Leak dissipation system shall consist of control board and A2L sensor certified to UL 60335-2-40
- System shall be designed for the life of the unit
- Dissipation system shall be automatic and disable compressors on 38AXQ condensing units
- Refrigerant leak sensor shall be installed in UL certified location and orientation. Sensor shall be self-correcting and resettable. Single use refrigerant leak sensor shall not be permitted
- Factory installed dissipation controller shall use onboard microprocessor and include:
  - Automatic reset after a dissipation event has occurred
  - Onboard LED with flash code to indicate current unit status and hardware failures
  - Depressible “Test” button to allow for a system test and recall/reset of leak detection history
  - 24V dry contact alarm terminal to allow for external notification of leak detection

- Dissipation control board shall be accessible via normal maintenance locations and LED shall be visible
- Dissipation system shall “Fail Safe” per UL requirements
- Dissipation shall allow smoke and building fire systems to override in case of event

Products

- Equipment
- Indoor mounted, draw-thru, packaged air-handling unit that can be used in a suspended horizontal configuration or a vertical configuration. Unit shall consist of a direct drive vane axial fan and motor assembly, pre-wired fan motor controller, factory-installed refrigerant metering devices (direct-expansion coil units), cooling coil, 2 in. (51 mm) disposable air filters, and condensate drain pans for vertical or horizontal configurations.
  - Cabinet shall be constructed of mill-galvanized steel.
  - Cabinet panels shall be fully insulated with 1/2 in. (12.7 mm) fire-retardant material.
  - Unit shall contain non-corroding condensate drain pans for both vertical and horizontal applications. Drain pans shall have connections on right and left sides of unit to facilitate field connection. Drain pans shall have the ability to be sloped toward the right or left side of the unit to prevent standing water from accumulating in pans.
  - Unit shall have factory-supplied 2 in. (51 mm) throwaway-type filters installed upstream from the cooling coil. Filter access shall be from either the right or left side of the unit.

B. Evaporator Fan and Motor with EcoBlue™ Technology:

- Direct Drive Evaporator fan motor:
  - Shall be an ECM motor design.
  - Shall be direct drive design for all static options.
  - Shall have permanently lubricated bearings.
  - Shall have inherent automatic-reset thermal overload protection.
  - Shall have slow ramp up to speed capabilities.
  - Shall require no fan/motor belts for operation, adjustments and or initial fan speed set up.
  - Shall be internally protected from electrical phase reversal and loss.
- Evaporator Fan:
  - Shall be easily set with dedicated selection switch and adjustment pot on unit control board
  - Shall provide two stage cooling capacity control, the indoor fan speed is automatically controlled to meet the code-compliant <66% low fan speed and 100% at full fan speed operation.
    - Blower fan shall be a Vane Axial fan design with fan assembly secured directly to ECM motor. Additional shafts, belts, pulleys/sheaves, and bearing blocks to drive fan shall not be permitted or necessary.
    - Additional variable frequency drive to control fan motor speed shall not be permitted or necessary. All speed control electronics must be onboard fan motor assembly.
  - Shall be constructed of a high impact composite material on stator, rotor and air inlet casing.
  - Shall be a patented / pending design with a corrosion resistant material and dynamically balanced.
  - Shall have slow ramp up to speed capabilities to help reduce sound and comfort issues typically

associated with single speed belt drive systems.

- Shall be a slide out design with removal of a few support brackets.

- Shall include an easily accessible Unit Control Board to conveniently and safely provide connection points for vital control functions such as: smoke detectors, phase monitor, economizer, thermostat, and low and high pressure switches. Controller shall also provide an intuitive means to adjust the indoor fan speed through a simple switch and pot adjustment design.

- Coils:
  - DX coil is 4-row and consists of copper tubes with sine-wave aluminum fins bonded to the tubes by mechanical expansion. Suction and liquid line connections or supply and discharge connections shall be made on the same side of the coil.
  - Direct-expansion coils shall feature factory installed thermostatic expansion valves (TXVs) for refrigerant control. The TXVs shall be Puron Advance® R-454B compatible and capable of external adjustment. Coil tubing shall be internally rifled to maximize heat transfer.
- Operating Characteristics:
  - When combined with matching 38AXQ condensing unit the system shall be capable of starting and running at ambient outdoor temperatures from 35°F (2°C) to 125°F (52°C) in cooling mode
  - Unit shall operate at +/–10% from rated voltage.
- Motor:
  - Fan motor of the size and electrical characteristics specified on the equipment schedule shall be factory supplied and installed.
  - Evaporator motors are designed specifically for Carrier and do not have conventional horsepower (hp) ratings listed on the motor nameplate. Motors are designed and qualified in the “air-over” location downstream of the cooling coil and carry a maximum continuous bhp rating that is the maximum application bhp rating for the motor; no “safety factors” above that rating may be applied.
- Special Features:
  - Alternate Motor and Drive:

    An alternate high-static motor shall be available to meet the airflow and external static pressure requirements specified on the equipment schedule.
  - External Paint:

    Where conditions require, units shall be painted with an American Sterling Gray finish.
  - Hot Water Coil:

    Coil shall be 2-row, U-bend coil with copper tubes and aluminum plate fins bonded to the tubes by mechanical expansion. Coil shall be mounted in a galvanized steel housing that shall be fastened to the unit’s fan deck for blow-thru heating operation. Coil shall have maximum working pressure of 150 psig.
  - Steam Distributing Coil:

    Coil shall consist of one row of copper tubes with aluminum plate fins, and shall have inner steam distributing tubes. Coil shall be mounted in a galvanized steel housing and shall be fastened to the unit’s fan deck for blow-thru heating operation. Coil shall have maximum working pressure of 20 psig at 260°F.
  - Electric Heaters:

Heaters for nominal 240v, 480v, or 575v, 3-phase, 60 Hz shall be factory-supplied for field installation as shown on the equipment drawings. Electric heat assembly shall be ETL (U.S.A.) and ETL, Canada, agency approved, and shall have single-point power wiring. Heater assembly shall include contactors with 24-v coils, power wiring, 24- v control wiring terminal blocks, and a hinged access panel. Electric heaters shall not be used with air discharge plenum.

- Air Discharge Plenum:

  Plenum shall be factory-supplied to provide free-blow air distribution for vertical floor-mounted units. A grille with moveable vanes for horizontal or vertical airflow adjustment shall be included. Plenum shall be field-assembled and field-installed on the unit's fan deck for blow-thru air distribution. Plenum shall not be used with electric heaters.

- Return-Air Grille:

  Grille shall be factory-supplied for field installation on the unit's return air opening.

- Unit Subbase:

  Subbase assembly shall be factory-supplied for field installation. Subbase shall elevate floor-mounted vertical units to provide access for correct condensate drain connection.

- Economizers:
  - Accessory Ultra Low Leak EconomizerONE. (Field-installed) Economizer for ventilation or "free" cooling shall be factory provided for field installation on either return air opening of air handler.
    - Integrated, gear driven opposing modulating blade design type capable of simultaneous economizer and compressor operation.
    - Damper blades shall be galvanized steel with composite gears. Plastic or composite blades on intake or return shall not be acceptable.
    - Shall include all hardware, actuator and controls to provide free cooling with outdoor air when temperature and/or humidity are below setpoints.
    - Shall be equipped with gear driven dampers for both the outdoor ventilation air and the return air for positive air stream control.
    - Ultra Low Leak design meets California Title 24 section 140.4 and ASHRAE90.1 requirements for 4 cfm per sq.ft. on the outside air dampers and 10 cfm per sq. ft. on the return dampers.
    - Economizer controller on EconomizerONE models shall be the Siemens POL224 that provides:
      - Combined minimum and DCV maximum damper position potentiometers with compressor staging relay.
      - Optional configuration via WLAN stick and Siemens Climatix™ smartphone app for easy setup.
      - Functions with solid-state analog enthalpy or dry bulb changeover control sensing.
      - LED indication for free cooling, sensor, and damper operation.
      - One-line LCD interface screen for setup, configuration and troubleshooting.
      - On-board Fault Detection and Diagnostics (FDD) that senses and alerts when the economizer is not operating properly, per California Title 24, ASHRAE 90.1 and IECC® .
      - Sensor failure loss of communication identification.
      - Capabilities for use with multiple-speed or single speed indoor fan systems.
      - Digital sensors: Dry bulb and Enthalpy.

- Overhead Suspension Package:

  Package shall include necessary brackets to support units in a horizontal ceiling installation.

- CO_2 Sensor:

  Sensor shall provide the ability to signal the economizer to open when the space CO_2 level exceeds the predetermined setpoint.

- Thermostat Controls:
  - Programmable multi-stage thermostat shall have 7-day clock, holiday scheduling, large backlit display, remote sensor capability, and Title 24 compliance.
  - Commercial Electronic Thermostat shall have 7-day time clock, auto-changeover, multi-stage capability, and large LCD (liquid crystal display) temperature display.

- Condensate Drain Trap:

  Trap shall have transparent, service-able design for easy cleaning. Kit shall include overflow shutoff switch and wiring harness for connection to an alarm if desired.

Commercial Packaged Air-Handling Unit

HVAC Guide Specifications

Size Range: **6,000 to 8,000 Cfm, Nominal Airflow, 15 to 25 Tons, Nominal Cooling**

Carrier Model Numbers: **40RLQ (Heat Pump Coil)**

General

- System Description
- Indoor, packaged air-handling unit for use in commercial heat pump split systems. Unit shall have a multi-position design and shall be capable of horizontal or vertical installation on a floor or in a ceiling, with or without ductwork. (Only vertical units are to be applied without ductwork.) Unit shall function as the indoor component of an air-to-air electric heat pump system.
  - Unit with direct-expansion coil shall use Puron Advance™ (R-454B) in a refrigerant circuit with a matching air-cooled condensing unit.
- Quality Assurance
  - Coils shall be designed and tested in accordance with ASHRAE 15 Safety Code for Mechanical Refrigeration (U.S.A.)
  - Unit shall be constructed in accordance with UL Standard 60335-1 and 60335-2-40, and shall carry the UL and UL, Canada, labels.
  - Unit insulation and adhesive shall comply with NFPA-90A (U.S.A.) requirements for flame spread and smoke generation.
  - Unit shall be manufactured in a facility registered to the ISO 9001 manufacturing quality standard.
  - Direct-expansion coils shall be burst and leak tested at 435 psi.
- Delivery And Storage
- Units shall be stored and handled per manufacturer's recommendations.
- Safeties

A. A2L Refrigerant Leak Dissipation System (Electromechanical)

- Leak dissipation system shall consist of control board and A2L sensor certified to UL 60335-2-40
- System shall be designed for the life of the unit
- Dissipation system shall be automatic and disable compressors on 38AX condensing units
- Refrigerant leak sensor shall be installed in UL certified location and orientation. Sensor shall be self-correcting and resettable. Single use refrigerant leak sensor shall not be permitted
- Factory installed dissipation controller shall use onboard microprocessor and include:
  - Automatic reset after a dissipation event has occurred
  - Onboard LED with flash code to indicate current unit status and hardware failures
  - Depressible "Test" button to allow for a system test and recall/reset of leak detection history
  - 24V dry contact alarm terminal to allow for external notification of leak detection
- Dissipation control board shall be accessible via normal maintenance locations and LED shall be visible
- Dissipation system shall "Fail Safe" per UL requirements
- Dissipation shall allow smoke and building fire systems to override in case of event

Products

- Equipment
- Indoor mounted, draw-thru, packaged air-handling unit that can be used in a suspended horizontal configuration or a vertical configuration. Unit shall consist of forward-curved belt-driven centrifugal fan(s), motor and drive assembly, pre-wired fan motor contactor, factory-installed refrigerant metering devices (direct-expansion coil units), cooling coil, 2 in. (51 mm) disposable air filters, and condensate drain pans for vertical or horizontal configurations.
  - Cabinet shall be constructed of mill-galvanized steel.
  - Cabinet panels shall be fully insulated with 1/2 in. (12.7 mm) fire-retardant material.
  - Unit shall contain non-corroding condensate drain pans for both vertical and horizontal applications. Drain pans shall have connections on right and left sides of unit to facilitate field connection. Drain pans shall have the ability to be sloped toward the right or left side of the unit to prevent standing water from accumulating in pans.
  - Unit shall have factory-supplied 2 in. (51 mm) throwaway-type filters installed upstream from the cooling coil. Filter access shall be from either the right or left side of the unit.
- Coils:
  - DX coil is 4-row and consists of copper tubes with sine-wave aluminum fins bonded to the tubes by mechanical expansion. Suction and liquid line connections or supply and discharge connections shall be made on the same side of the coil.
  - Direct-expansion coils shall feature factory installed thermostatic expansion valves (TXVs) for refrigerant control. The TXVs shall be Puron Advance® R-454B compatible and capable of external adjustment. Coil tubing shall be internally rifled to maximize heat transfer.
- Operating Characteristics:
  - When combined with matching 38AXQ condensing unit the system shall be capable of starting and running at ambient outdoor temperatures from 35°F (2°C) to 125°F (52°C) in cooling mode.
  - Unit shall operate at +/–10% from rated voltage.
- Motor:
  - Fan motor of the size and electrical characteristics specified on the equipment schedule shall be factory supplied and installed.
  - Motors rated at 1.3 through 3.7 hp shall have inherent thermal overload protection. Motors rated at 5 hp shall be protected by a circuit breaker.
  - Evaporator-fan motor shall have permanently lubricated, sealed bearings and inherent automatic-reset thermal overload protection or manual reset calibrated circuit breakers. Evaporator motors are designed specifically for Carrier and do not have conventional horsepower (hp) ratings listed on the motor nameplate. Motors are designed and qualified in the “air-over” location downstream of the cooling coil and carry a maximum continuous bhp rating that is the maximum application bhp rating for the motor; no “safety factors” above that rating may be applied.
  - All evaporator-fan motors 5 hp and larger shall meet the minimum efficiency requirements as established by the Energy Policy Act of 1992 (EPACT), effective October 24, 1997.
- Control Box:

- Shall include an easily accessible Unit Control Board to conveniently and safely provide connection points for vital control functions such as: smoke detectors, phase monitor, economizer, thermostat, and low and high pressure switches. Controller shall also provide an intuitive means to adjust the indoor fan speed through a simple switch and pot adjustment design.

- Staged Air Volume System (SAV™) for 2-stage cooling models (standard):

  Evaporator fan motor.

  - Shall have permanently lubricated bearings.
  - Shall have a maximum continuous bhp rating for continuous duty operation; no safety factors above that rating.
  - Shall be Variable Frequency duty and 2-speed control.
  - Shall contain motor shaft grounding ring to prevent electrical bearing fluting damage by safely diverting harmful shaft voltages and bearing currents to ground.

  2. Variable Frequency Drive (VFD) Standard:

     a. Shall be installed inside the unit cabinet, mounted, wired and tested.

     b. Shall contain Electromagnetic Interference (EMI) frequency protection.

     c. Insulated Gate Bi-Polar Transistors (IGBT) used to produce the output pulse width modulated (PWM) wave-form, allowing for quiet motor operation.

     d. Self diagnostics with fault and power code LED indicator. Field accessory Display Kit available for further diagnostics and special setup applications.

     e. RS485 capability standard.

     f. Electronic thermal overload protection.

     g. 5% swinging chokes for harmonic reduction and improved power factor.

     h. All printed circuit boards shall be con-formal coated.

- Special Features:
  - Alternate Motor and Drive:

    An alternate motor and/or medium-static or high-static drive shall be available to meet the airflow and external static pressure requirements specified on the equipment schedule.

  - External Paint:

    Where conditions require, units shall be painted with an American Sterling Gray finish.

  - Hot Water Coil:

    Coil shall be 2-row, U-bend coil with copper tubes and aluminum plate fins bonded to the tubes by mechanical expansion. Coil shall be mounted in a galvanized steel housing that shall be fastened to the unit's fan deck for blow-thru heating operation. Coil shall have maximum working pressure of 150 psig.

  - Steam Distributing Coil:

    Coil shall consist of one row of copper tubes with aluminum plate fins, and shall have inner steam distributing tubes. Coil shall be mounted in a galvanized steel housing and shall be fastened to the unit's fan deck for blow-thru heating operation. Coil shall have maximum working pressure of 20 psig

at 260°F.

- Electric Heaters:

  Heaters for nominal 240v, 480v, or 575v, 3-phase, 60 Hz shall be factory-supplied for field installation as shown on the equipment drawings. Electric heat assembly shall be UL (U.S.A.) and UL, Canada, agency approved, and shall have single-point power wiring. Heater assembly shall include contactors with 24-v coils, power wiring, 24- v control wiring terminal blocks, and a hinged access panel. Electric heaters shall not be used with air discharge plenum.

- Air Discharge Plenum:

  Plenum shall be factory-supplied to provide free-blow air distribution for vertical floor-mounted units. A grille with moveable vanes for horizontal or vertical airflow adjustment shall be included. Plenum shall be field-assembled and field-installed on the unit's fan deck for blow-thru air distribution. Plenum shall not be used with electric heaters.

- Return-Air Grille:

  Grille shall be factory-supplied for field installation on the unit's return air opening.

- Unit Subbase:

  Subbase assembly shall be factory-supplied for field installation. Subbase shall elevate floor-mounted vertical units to provide access for correct condensate drain connection.

- Economizers:
  - Accessory Ultra Low Leak EconomizerONE. (Field-installed) Economizer for ventilation or "free" cooling shall be factory provided for field installation on either return air opening of air handler.
    - Integrated, gear driven opposing modulating blade design type capable of simultaneous economizer and compressor operation.
    - Damper blades shall be galvanized steel with composite gears. Plastic or composite blades on intake or return shall not be acceptable.
    - Shall include all hardware, actuator and controls to provide free cooling with outdoor air when temperature and/or humidity are below setpoints.
    - Shall be equipped with gear driven dampers for both the outdoor ventilation air and the return air for positive air stream control.
    - Ultra Low Leak design meets California Title 24 section 140.4 and ASHRAE90.1 requirements for 4 cfm per sq.ft. on the outside air dampers and 10 cfm per sq. ft. on the return dampers.
    - Economizer controller on EconomizerONE models shall be the Siemens POL224 that provides:
      - Combined minimum and DCV maximum damper position potentiometers with compressor staging relay.
      - Optional configuration via WLAN stick and Siemens Climatix™ smartphone app for easy setup.
      - Functions with solid-state analog enthalpy or dry bulb changeover control sensing.
      - LED indication for free cooling, sensor, and damper operation.
      - One-line LCD interface screen for setup, configuration and troubleshooting.
      - On-board Fault Detection and Diagnostics (FDD) that senses and alerts when the economizer is not operating properly, per California Title 24, ASHRAE 90.1 and IECC® .
      - Sensor failure loss of communication identification.
      - Capabilities for use with multiple-speed or single speed indoor fan systems.
      - Digital sensors: Dry bulb and Enthalpy.
- Overhead Suspension Package:

Package shall include necessary brackets to support units in a horizontal ceiling installation.

- CO_2 Sensor:

  Sensor shall provide the ability to signal the economizer to open when the space CO_2 level exceeds the predetermined setpoint.

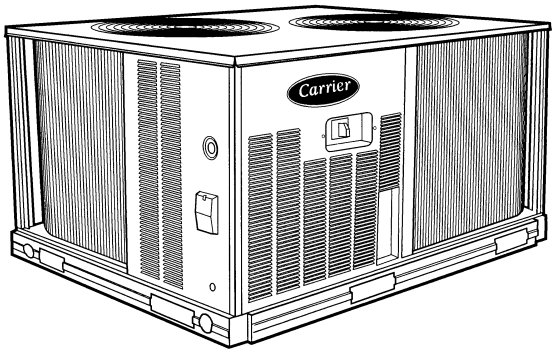
- Thermostat Controls:
  - Programmable multi-stage thermostat shall have 7-day clock, holiday scheduling, large backlit display, remote sensor capability, and Title 24 compliance.
  - Commercial Electronic Thermostat shall have 7-day time clock, auto-changeover, multi-stage capability, and large LCD (liquid crystal display) temperature display.
- Condensate Drain Trap:

  Trap shall have transparent, service-able design for easy cleaning. Kit shall include overflow shutoff switch and wiring harness for connection to an alarm if desired.

- Display Kit for Variable Frequency Drive:
  - Kit allows the ability to access the VFD con-troller programs to provide special setup capabilities and diagnostics.
  - Kit contains display module and communications cable.
  - Display Kit can be permanently installed in the unit or used on any SAV system VFD controller as needed.

COMMERCIAL SPLIT SYSTEMS
38AXQ Single Circuit, Two Stage 6, 7.5, 10 Ton
AIR-COOLED HEAT PUMP UNITS with PURON ADVANCE® REFRIGERANT

These commercial quality air-cooled heat pump units are easily connected by refrigerant lines and low voltage control wiring to matching Carrier packaged air-handling units. They are ideal for new construction or renovation applications where quality and performance are required. All units use Carrier's new, lower global warming potential Puron Advance™ (R-454B) refrigerant. Puron Advance™ represents a 75% reduction in refrigerant GWP over legacy Puron™ (R-410A) models.



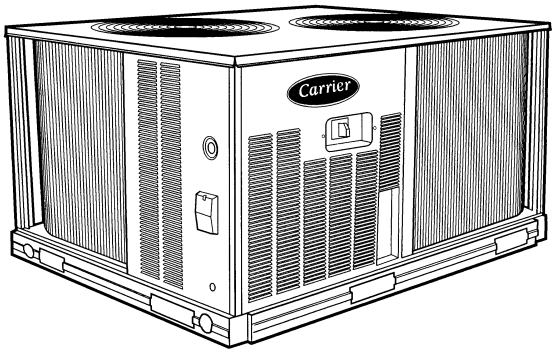
Use of the AHRI Certified TM Mark indicates a manufacturer's participation in the program For verification of certification for individual products, go to www.ahridirectory.org.

BASE UNIT STANDARD FEATURES:

- Puron Advance® (R-454B) refrigerant
- Single circuit two-stage cooling capacity control on 07-12 models
- All model sizes utilize copper tube/aluminum plate fin condenser coils.
- Standard one-year unit warranty and 5-year compressor plan
- Cooling operating range up to 125 F (52 C) and down to 35 F (2 C)
- Brass suction and liquid line service valves
- Fully Hermetic Scroll compressor with crankcase heater and suction line accumulator
- New terminal board facilitating simple safety circuit troubleshooting and simplified control box arrangement. 24-volt 75va control system
- High and low pressure switches
- Comfort Alert™ Diagnostic Board:
  - LED Go-No-Go and fault code
  - Built in time guard anti-short cycle
  - Phase protection
  - Fault code retention logic
  - Low voltage compressor contactor protector
- Rated in accordance with AHRI Standards 340/360
- Designed in accordance with Underwriters' Laboratories Standard UL 60335-1 and UL 60335-2-40
- Listed by UL and CUL-Canada
- Full perimeter base rails with built-in rigging adapters and fork truck slots
- Pre-painted exterior panels and primer-coated interior panels tested to 500 hours salt spray protection
- Direct drive permanently lubricated condenser fan motors
- All units factory run tested
- Compressors mounted on independent vibration isolators
- High capacity filter drier is supplied for each circuit. Field installation is required.

COMMERCIAL SPLIT SYSTEMS
38AXQ Dual Circuit, Three Stage 15 – 20 Ton
AIR-COOLED HEAT PUMP UNITS with PURON ADVANCE® REFRIGERANT

These commercial quality air-cooled heat pump units are easily connected by refrigerant lines and low voltage control wiring to matching Carrier packaged air-handling units. They are ideal for new construction or renovation applications where quality and performance are required. All units use Carrier's new, lower global warming potential Puron Advance™ (R-454B) refrigerant. Puron Advance™ represents a 75% reduction in refrigerant GWP over legacy Puron™ (R-410A) models.



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BASE UNIT STANDARD FEATURES:

- **Dual circuit, three stage cooling capacity on 16-25 size models**
- Puron Advance® (R-454B) refrigerant
- All models utilize copper tube/aluminum plate fin condenser coils
- Standard one-year unit warranty and 5-year compressor plan
- Cooling operating range up to 125 F (52 C) and down to 35 F (2 C)
- Brass suction and liquid line service valves
- Fully Hermetic Scroll compressor with crankcase heater
- New terminal board facilitating simple safety circuit troubleshooting and simplified control box arrangement. 24-volt 75va control system
- High and low pressure switches.
- Comfort Alert™ Diagnostic Board:
  - LED Go-No-Go and fault code
  - Built in time guard anti-short cycle
  - Phase protection
  - Fault code retention logic
  - Low voltage compressor contactor protector
- Rated in accordance with AHRI Standards 340/360
- Designed in accordance with Underwriters' Laboratories Standard UL 60335-1 and UL 60335-2-40
- Listed by UL and CUL-Canada
- Full perimeter base rails with built-in rigging adapters and fork truck slots
- Pre-painted exterior panels and primer-coated interior panels tested to 500 hours salt spray protection
- Direct drive permanently lubricated condenser fan motors

- All units factory run tested
- Compressors mounted on independent vibration isolators
- High capacity filter drier is supplied for each circuit. Field installation is required.

Gemini

COMMERCIAL SPLIT SYSTEMS
40RLA DIRECT EXPANSION PACKAGED AIR-HANDLING UNITS
6 – 10 Ton
with PURON ADVANCE® REFRIGERANT

The 40RLQ units are complete packaged heat pump air-handling units offering the ultimate in installation ease and flexibility. Combined with Carrier 38AXQ Series heat pump condensing units, these packaged air handling units offer system capacities from 6 up to 10 tons. These units may be installed in either the vertical or horizontal configuration with no modifications. Powerful fan systems allow easy adaptation to existing ductwork. Heating capability is provided by electric heaters, hot water coils or steam coils. All units use Carrier's new, lower global warming potential Puron Advance™ (R-454B) refrigerant. Puron Advance™ represents a 75% reduction in refrigerant GWP over legacy Puron™ (R-410A) models.



40RLQ 6 TO 10 TON CHASSIS SHOWN

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ASHRAE 90.1 COMPLIANT

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BASE UNIT STANDARD FEATURES:

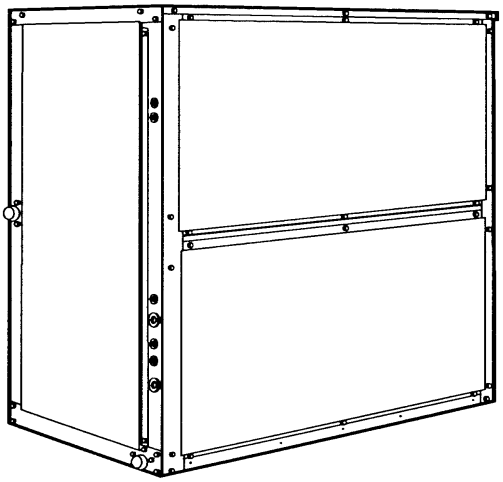
- Puron Advance® (R-454B) refrigerant
- New - A2L leak detection and dissipation system factory installed standard
- Leak system ensures unit and occupant safety during operation and includes an alarm relay for optional use
- Onboard recallable leak detection history for easier troubleshooting and service
- Direct Drive – EcoBlue™ Technology Indoor fan system uses Vane Axial fan design and electronically commutated motor. Indoor fan motor delivers Staged Air Volume (SAV) fan speed control
- New Unit Control Board with intuitive quick fan speed adjustment
- Two sloped condensate pans on each unit for horizontal or vertical applications.
- 2 inch filters.

- Refrigerant metering devices (TXV's) -- factory installed.
- Easy maintenance -- removal of a single panel allows access to virtually all components.
- Single refrigerant circuit on 07 and 08 models, dual refrigerant circuit on size 12 models. Dual circuit models can be field modified for use on single circuit condensing units.
- Single blower on 07 to 12 sizes
- 24-volt terminal block for control wiring connection.
- Factory installed Staged Air Volume (SAV™) system with 2-speed indoor fan system
- Standard one year warranty.
- Designed in accordance with Underwriters' Laboratories Standard UL 60335-1 and UL 60335-2-40
- Listed by UL and CUL-Canada

Gemini

COMMERCIAL SPLIT SYSTEMS
40RLQ DIRECT EXPANSION PACKAGED AIR-HANDLING UNITS
15 – 20 Ton
with PURON ADVANCE® REFRIGERANT

The 40RLQ units are complete packaged heat pump air-handling units offering the ultimate in installation ease and flexibility. Combined with Carrier 38AXQ Series heat pump condensing units these packaged air handling units offer system capacities from 15 up to 20 Tons. These units may be installed in either the vertical or horizontal configuration with no modifications. Powerful fan systems allow easy adaptation to existing ductwork. Heating capability is provided by electric heaters, hot water coils or steam coils. All units use Carrier's new, lower global warming potential Puron Advance™ (R-454B) refrigerant. Puron Advance™ represents a 75% reduction in refrigerant GWP over legacy Puron™ (R-410A) models.



40RLQ 15 TO 20 Ton CHASSIS SHOWN

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ASHRAE 90.1 COMPLIANT

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BASE UNIT STANDARD FEATURES:

- Puron Advance® (R-454B) refrigerant
- New - A2L leak detection and dissipation system factory installed standard
- Leak system ensures unit and occupant safety during operation and includes an alarm relay for optional use
- Onboard recallable leak detection history for easier troubleshooting and service
- Two sloped condensate pans on each unit for horizontal or vertical applications.
- 2 inch filters.
- Powerful belt-driven forward curved fans.
- Staged Air Volume (SAV™) system with 2-speed indoor fan VFD controller standard on all size 16-25 models
- Factory-installed fan motor and contactor.
- Refrigerant metering devices (TXV's) -- factory installed.
- Easy maintenance -- removal of a single panel allows access to virtually all components.
- Dual refrigerant circuit on 16-25 models. Dual circuit models can be field modified for use on single circuit condensing units.
- Dual blower on 16 to 25 sizes.
- 24-volt terminal block for control wiring connection.
- New - Unit Control Board (UCB) standard to allow for easy set up
- Standard one year warranty.
- Designed in accordance with Underwriters' Laboratories Standard UL 60335-1 and UL 60335-2-40
- Listed by UL and CUL-Canada